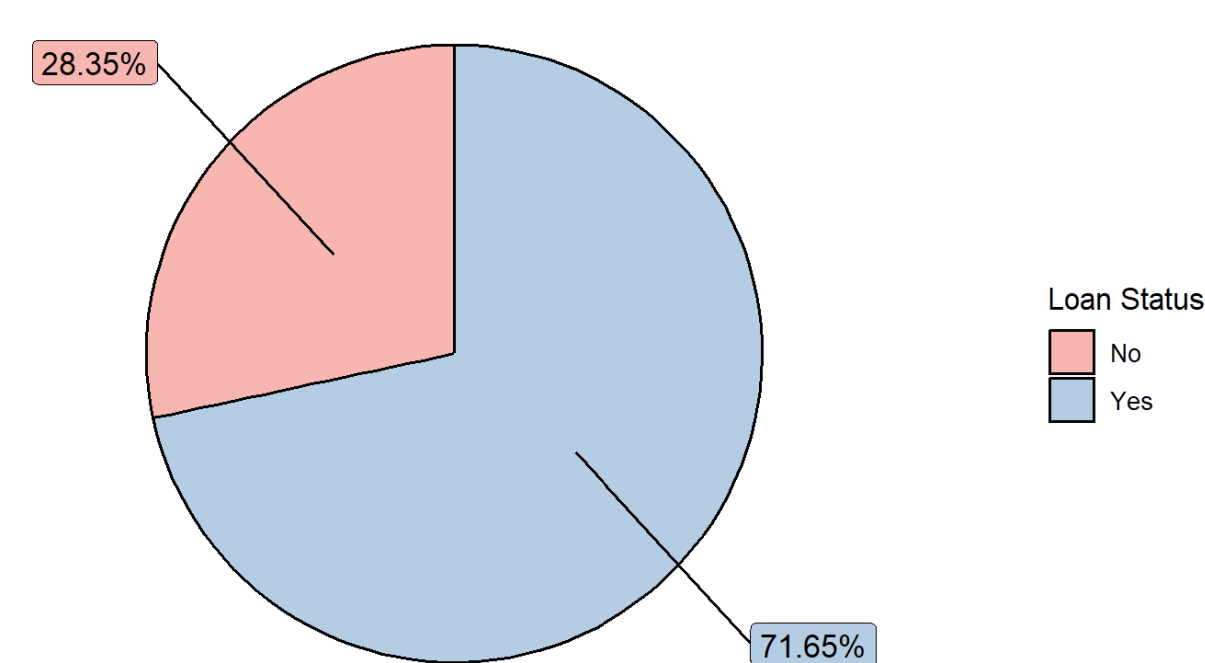


Classification analysis for Loan Status



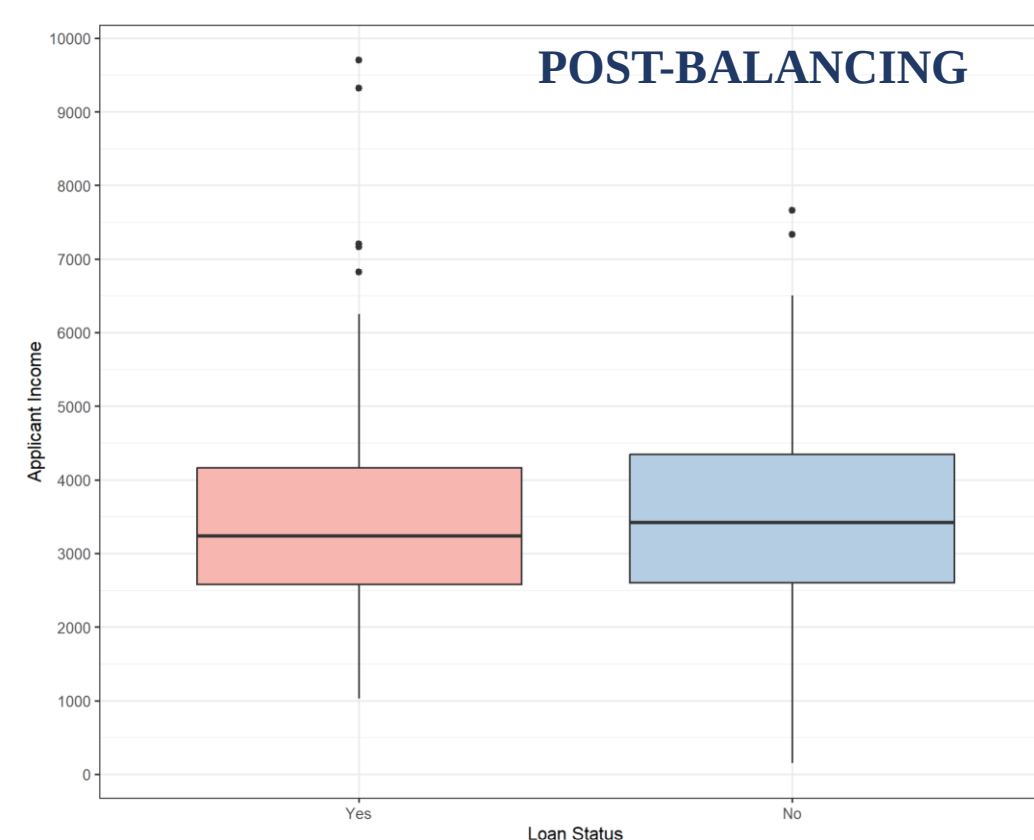
The analysis has the purpose of the analysis is correctly predicting the Loan Status based on the dataset's variables, which may be not all relevant. The variables indicate the Gender, the Marital status, the number of dependents, the educational level, the type of work, the property area, the guidelines satisfaction in the past. The dataset includes also the applicant and co-applicant income, the loan amount and the loan term.

LOAN STATUS

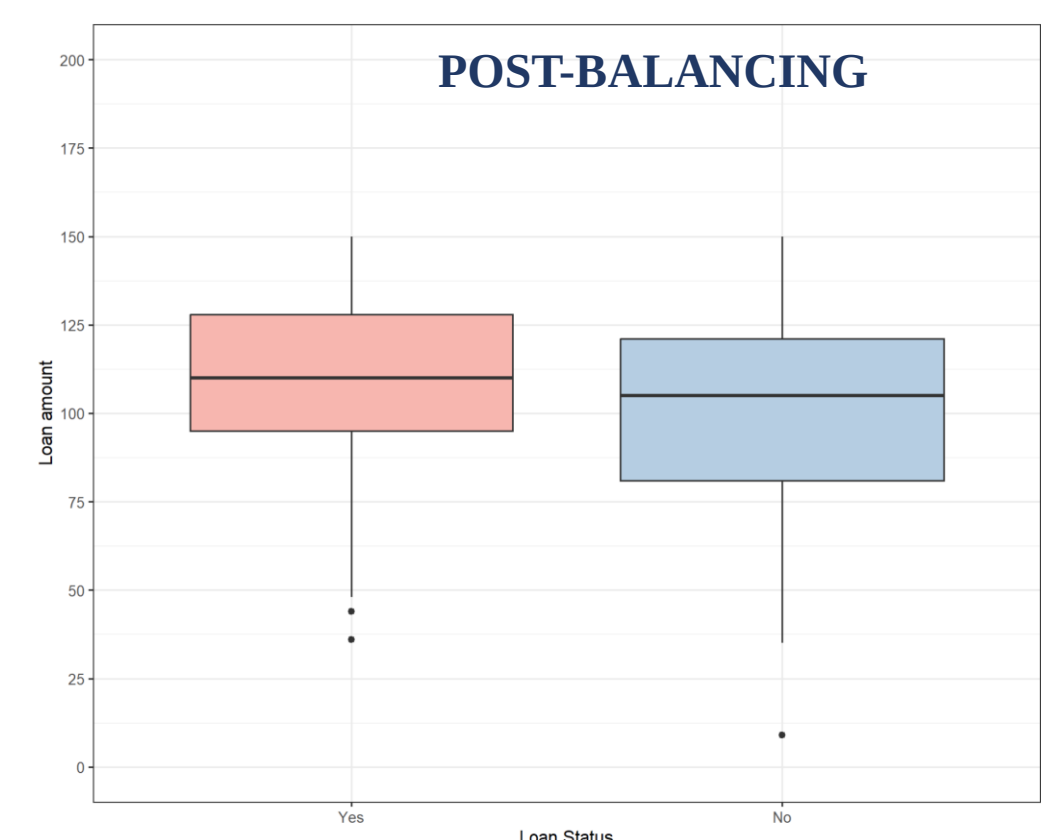


Dependents	
0	125
1	25
2	26
3+	10

APPLICANT INCOME



LOAN AMOUNT



Urban	Semiurban	Rural
57	72	57

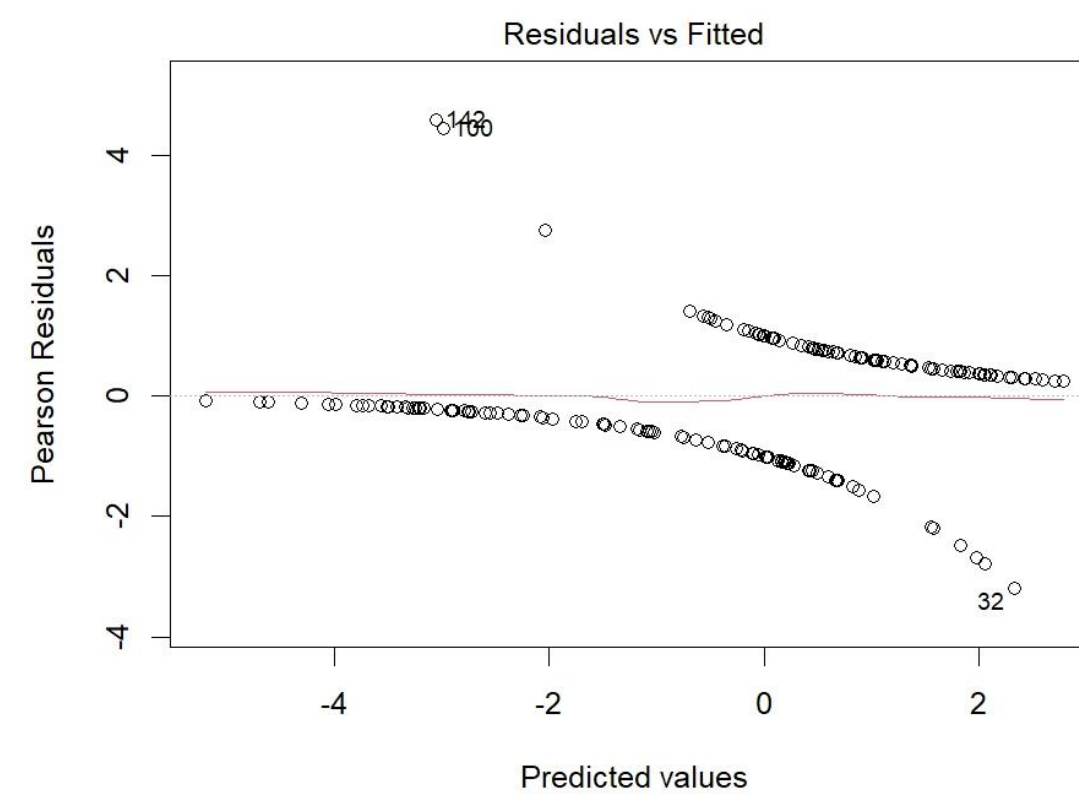
	Male	Married	Graduate	Self employed	Guidelines met
Yes	143	101	134	19	140
No	43	85	52	167	46

PREDICTION MODELS ON BALANCED DATASET WITH ALL THE VARIABLES

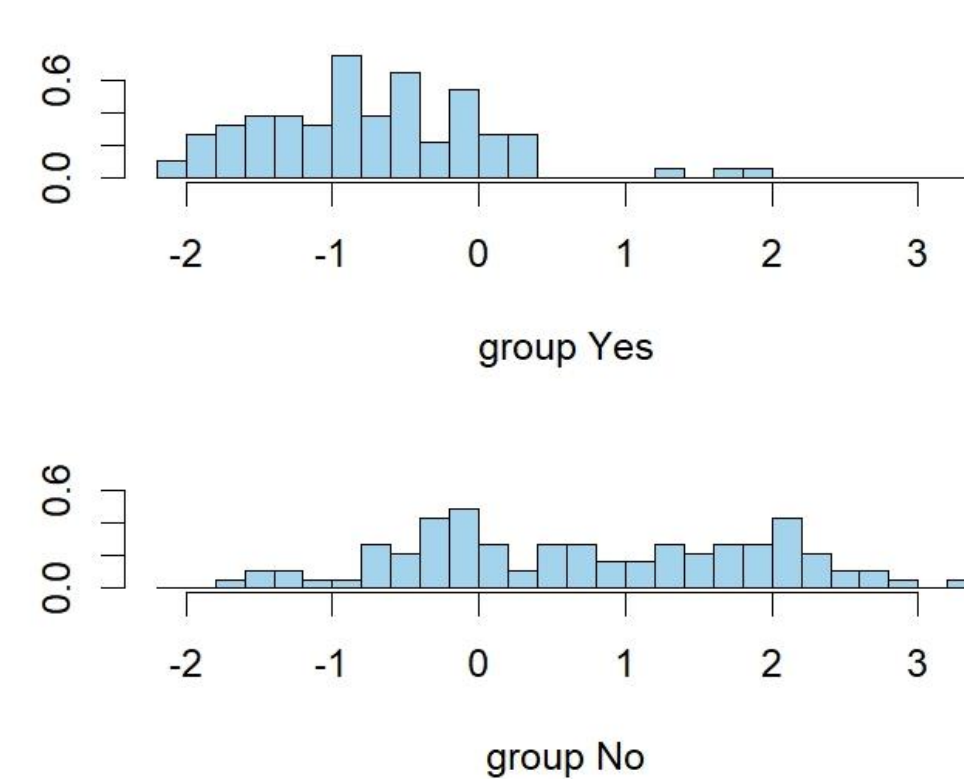
CONFUSION MATRICES' VALUES

	Logistic	LDA	QDA	Naive Bayes
Accuracy	0.6383	0.6809	0.6809	0.7660
AccuracyPValue	0.0935	0.0276	0.0276	0.0008
Kappa	0.2834	0.3711	0.3711	0.5413
Sensitivity	0.7273	0.8182	0.8182	0.9545
Specificity	0.5600	0.5600	0.5600	0.6000

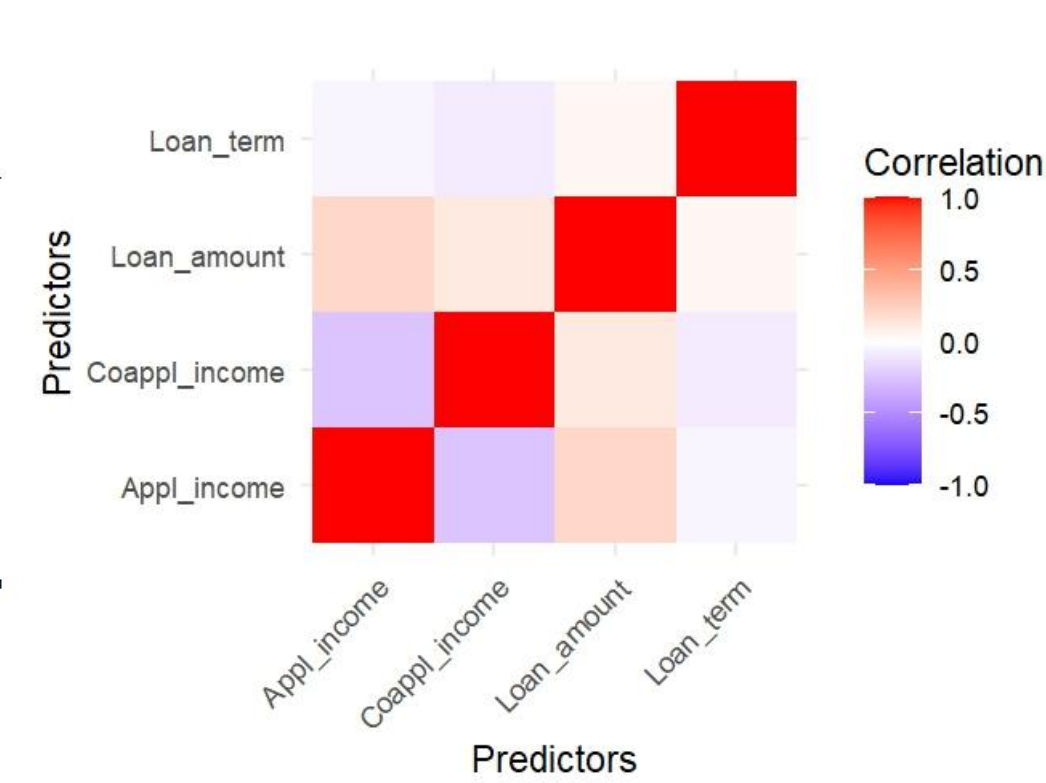
NO LINEAR RELATIONSHIP



NO CLASSES SEPARATION



NO CORRELATION



SUBSET SELECTION METHODS WITH BINARY RESPONSE VARIABLE

BEST SUBSET SELECTION

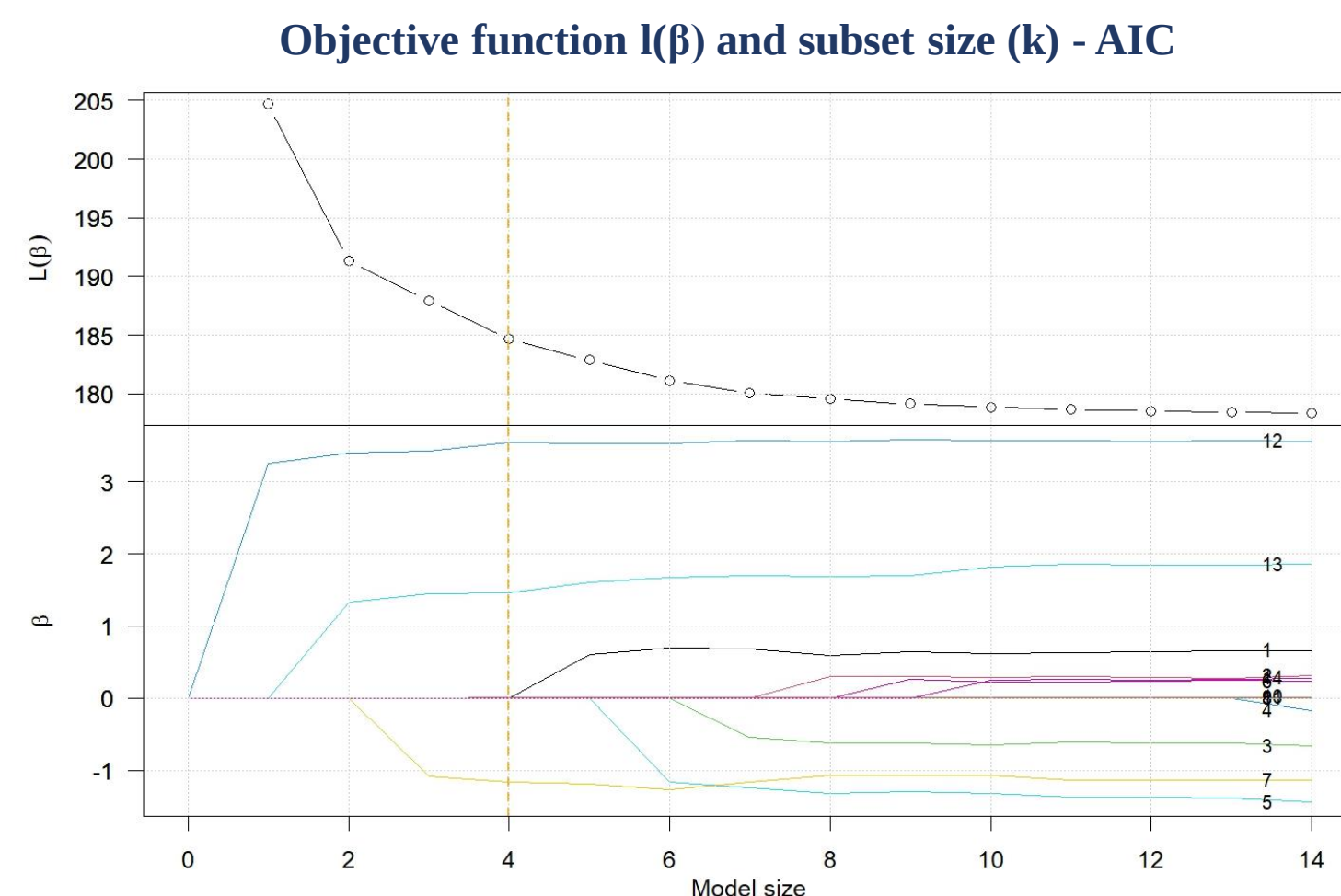
```
# Creating a matrix containing the predictor variables
predictive_matrix <- model.matrix(~. -Loan_status -loan_status.rec,
                                  SLproject_balanced)[, -1]

head(predictive_matrix)
y <- as.matrix(SLproject_balanced$Loan_status)
# SEQUENTIAL = it is similar to the mixed strategy
# Perform best subset selection with bess
fit.seq <- bess(x = predictive_matrix, y = y,
               family = "binomial", method = "sequential",
               weights = rep(1,nrow(predictive_matrix)))

# We found k variables according AIC
K.opt.aic <- which.min(fit.seq$AIC)
coef(fit.seq)[, K.opt.aic][which(coef(fit.seq)[, K.opt.aic] != 0)]
plot(fit.seq, type = "both", breaks = TRUE, K = K.opt.aic)

# We found k variables according BIC
K.opt.bic <- which.min(fit.seq$BIC)
coef(fit.seq)[, K.opt.bic][which(coef(fit.seq)[, K.opt.bic] != 0)]
plot(fit.seq, type = "both", breaks = TRUE, K = K.opt.bic)

# We found k variables according EBIC
K.opt.ebic <- which.min(fit.seq$EBIC)
coef(fit.seq)[, K.opt.ebic][which(coef(fit.seq)[, K.opt.ebic] != 0)]
plot(fit.seq, type = "both", breaks = TRUE, K = K.opt.ebic)
```



RIDGE REGRESSION

```
grid <- 10^seq(-2, 5, 0.01)
ridge.mod <- glmnet(predictive_matrix, y, family="binomial",
                    alpha = 0, lambda = grid)
```

LASSO REGRESSION

```
lasso.mod <- glmnet(predictive_matrix, y, family="binomial",
                    alpha = 1, lambda = grid)
```

STEPWISE SELECTION

```
# Logistic regression model with all variables
model <- glm(Loan_status ~ . -Loan_status - loan_status.rec,
            data = SLproject_balanced, family = "binomial")
# Variables selection with stepAIC
selected_model <- stepAIC(model, direction = "both")
```

Selected variables are Self_employed, Loan_amount, Guidelines_met, Prop_areaSemiurban

PREDICTION MODELS ON BALANCED DATASET WITH SELECTED VARIABLES

LOGISTIC REGRESSION

Accuracy : 0.7234
95% CI : (0.5736, 0.8438)
No Information Rate : 0.5319
P-Value [Acc > NIR] : 0.005762

Kappa : 0.455

Mcnemar's Test P-Value : 0.096092

Sensitivity : 0.8636
Specificity : 0.6000
Pos Pred Value : 0.6552
Neg Pred Value : 0.8333
Prevalence : 0.4681
Detection Rate : 0.4043
Detection Prevalence : 0.6170
Balanced Accuracy : 0.7318

'Positive' Class : Yes

LDA

Accuracy : 0.7021
95% CI : (0.5511, 0.8266)
No Information Rate : 0.5319
P-Value [Acc > NIR] : 0.01319

Kappa : 0.4146

Mcnemar's Test P-Value : 0.06137

Sensitivity : 0.8636
Specificity : 0.5600
Pos Pred value : 0.6332
Neg Pred Value : 0.8235
Prevalence : 0.4681
Detection Rate : 0.4043
Detection Prevalence : 0.6383
Balanced Accuracy : 0.7118

'Positive' Class : Yes

QDA

Accuracy : 0.7021
95% CI : (0.5511, 0.8266)
No Information Rate : 0.5319
P-Value [Acc > NIR] : 0.01319

Kappa : 0.4177

Mcnemar's Test P-Value : 0.01616

Sensitivity : 0.9091
Specificity : 0.5200
Pos Pred Value : 0.6250
Neg Pred Value : 0.8667
Prevalence : 0.4681
Detection Rate : 0.4255
Detection Prevalence : 0.6809
Balanced Accuracy : 0.7145

'Positive' Class : Yes

NAÏVE BAYES

Accuracy : 0.6809
95% CI : (0.5288, 0.8091)
No Information Rate : 0.5319
P-Value [Acc > NIR] : 0.02757

Kappa : 0.3744

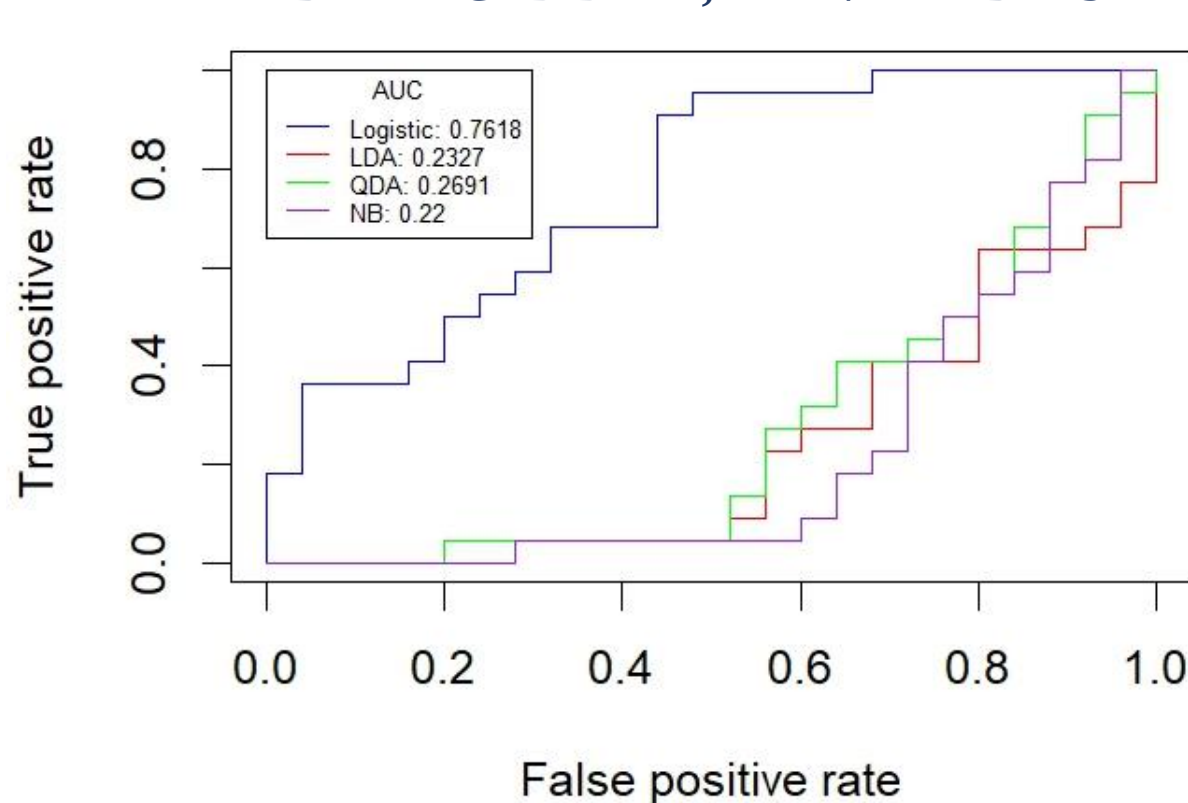
Mcnemar's Test P-Value : 0.03887

Sensitivity : 0.8636
Specificity : 0.5200
Pos Pred Value : 0.6129
Neg Pred Value : 0.8125
Prevalence : 0.4681
Detection Rate : 0.4043
Detection Prevalence : 0.6596
Balanced Accuracy : 0.6918

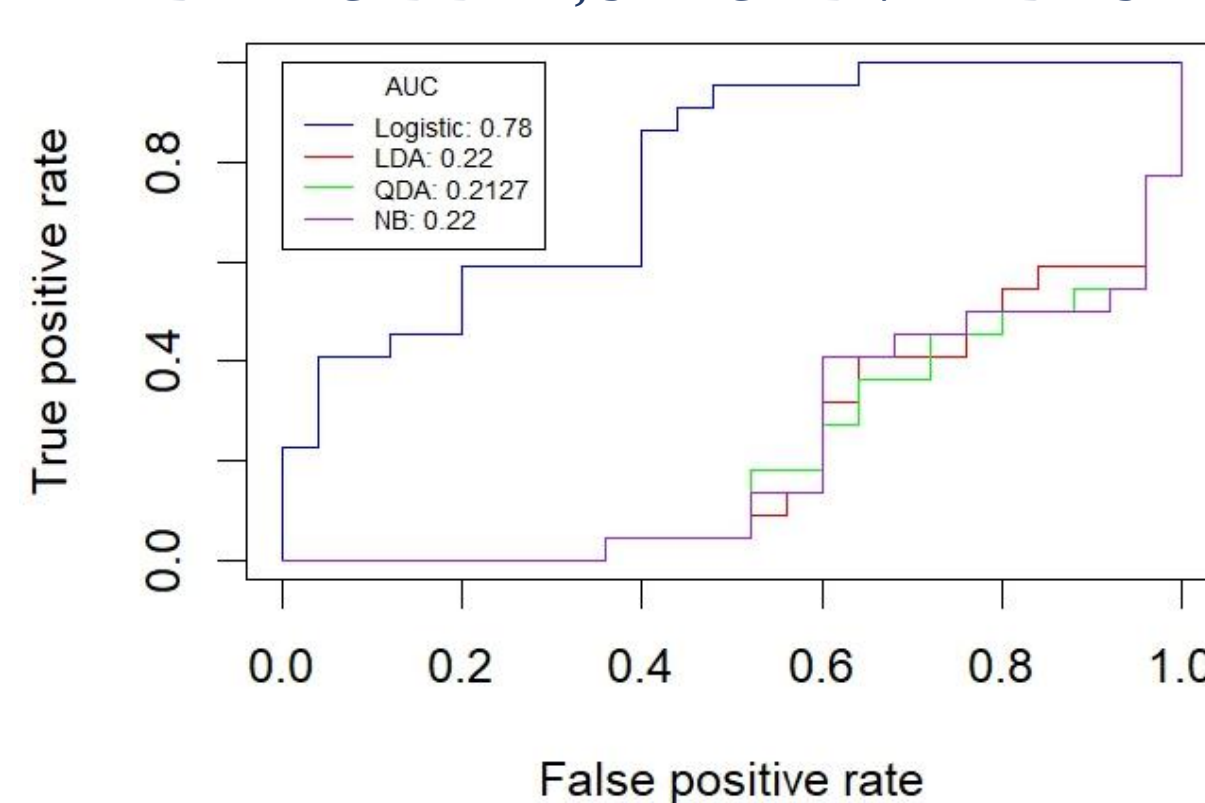
'Positive' Class : Yes

ROC CURVE

BALANCED DATA, ALL VARIABLES



BALANCED DATA, SELECTED VARIABLES



CONCLUSIONS

The best model seems to be the Naive Bayes. however, its AUC is too low. It is due to the low heterogeneity of qualitative variables and the fact that quantitative variables do not help predict in predicting the loan status.